Tilecal Calibration, Data Quality, Performance and Processing

Quantization and FPGA Synthesis for TileCal Energy Reconstruction Neural Network

77 channels placed and routed on one KU115 at 280 MHz

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Where the project stands

01

Optimization of parameters

How big does the network need to be?

02

Teacher/student

Getting under 100 parameters.

03

Quantization

How few bits does it survive?

04

Synthesis

Does it fit and does it meet the rate?

Blocks 1-3 closed. Block 4 now has routed numbers.

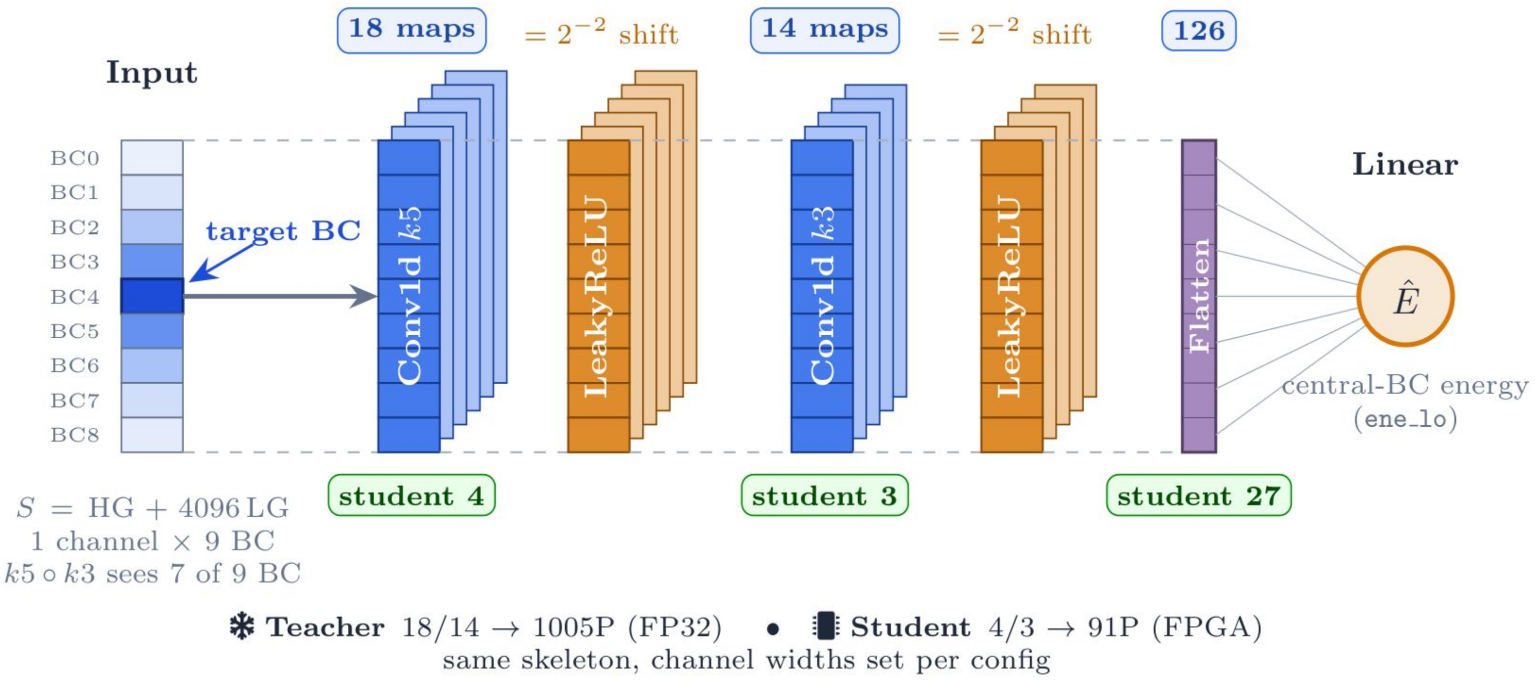
NEXT

Pedestal correction then Athena.

Recap, the model and how we read resolution

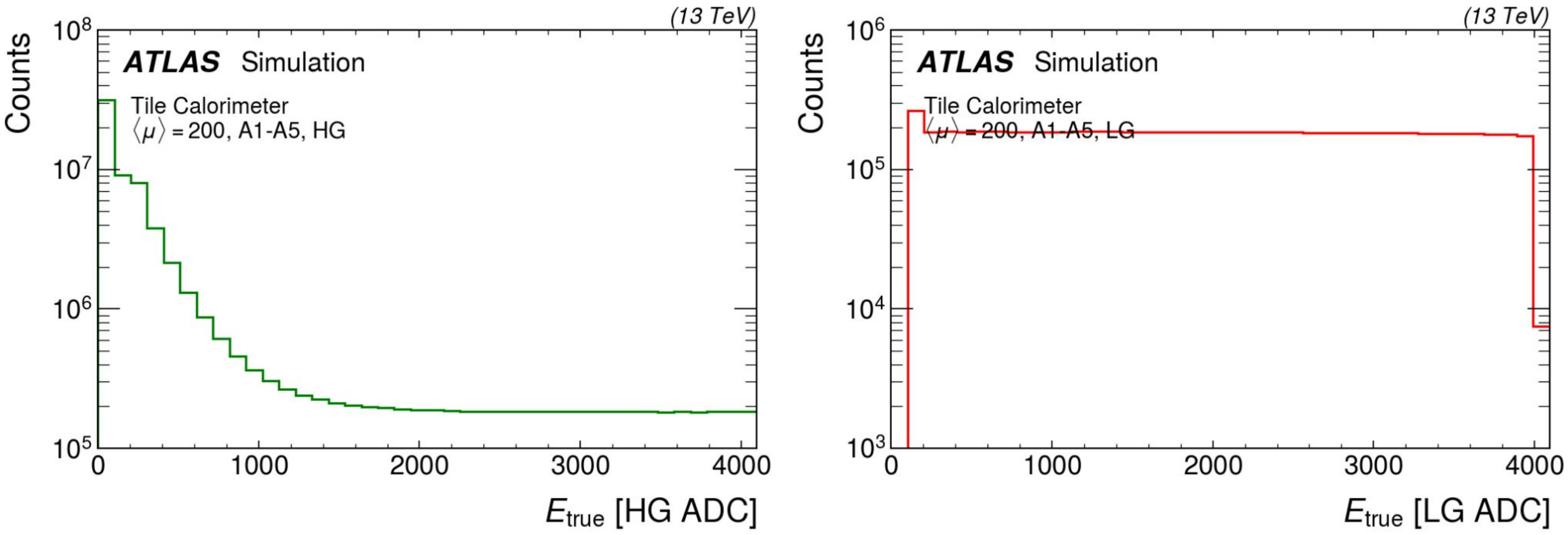
- 1D-CNN on a 9-bunch-crossing window. Conv1d(1->c1,k5) -> LeakyReLU(0.25) -> Conv1d(c1->c2,k3) -> LeakyReLU(0.25) -> Linear(9*c2 -> 1).
- One concatenated dual-gain input `S = HG + 4096*LG`, one energy output. No separate per-gain model.
- Target is the central bunch crossing. LeakyReLU(0.25) is a 2^-2 bit shift, free in firmware.
- Teacher 18/14 channels, 1005 parameters. Student 4/3, 91 parameters.

Resolution, two numbers, used throughout. global = std over all test events. band = mean of the per-energy-bin sigma. We quote the band.



The dataset

- 230,000 consecutive bunch crossings, mu = 200 minimum bias, 5% flat signal injection biased to high gain. Cells A1 to A5.
- 71,061,518 windows kept after filters. Split 70 / 15 / 15.
- Every number here is on the same test split of 10,659,229 events, 9,597,174 high gain and 1,062,055 low gain.
- 90% of windows sit in the high-gain region. Median true energy 3.5 LG ADC.



Public plot is cell A1, ours is A1-A5

Parameter optimization and teacher/student conclusions

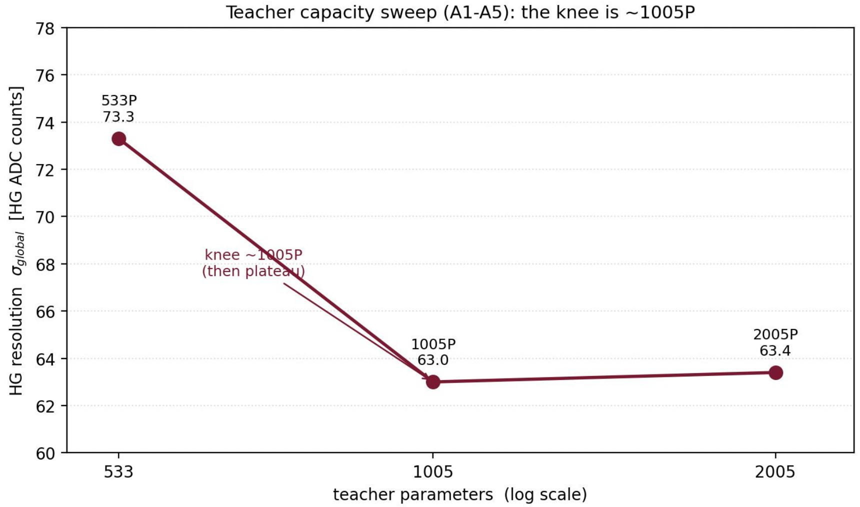
Model	HG global (HG ADC)	HG band (HG ADC)	LG band (LG ADC)
teacher 533 P	73.3	101.8	2.99
teacher 1005 P	63.0	85.8	2.70
teacher 2005 P	63.4	85.5	2.68
student 91 P, from scratch	105.9	164.8	4.72
student 91 P, distilled	**95.3**	**127.0**	**3.43**

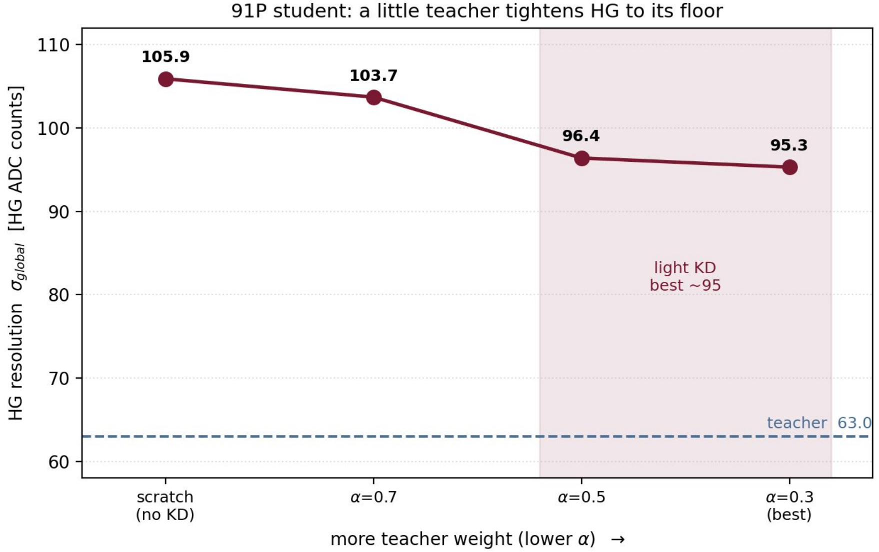
- Knee is near 1000 parameters. 533 to 1005 tightens the band 16%, 1005 to 2005 does nothing.
- Distillation buys 23% of band at fixed size, 164.8 to 127.0 HG ADC, best at alpha = 0.3.

Dead ends, all tie or trail alpha = 0.3. k-teacher, feature distillation, teacher-assistant, scale-fixed matching, relative-resolution loss.

Caveat. 533 P looked like a floor too, until cells were added. So the 2005 P plateau is a capacity floor or a data ceiling and these runs cannot tell which.

The 91 P distilled student is what goes to firmware.





The bit-width frontier

- Brevitas QAT from the float student. Per-tensor power-of-two quantizers.

Activations across, weights down. HG band in HG ADC

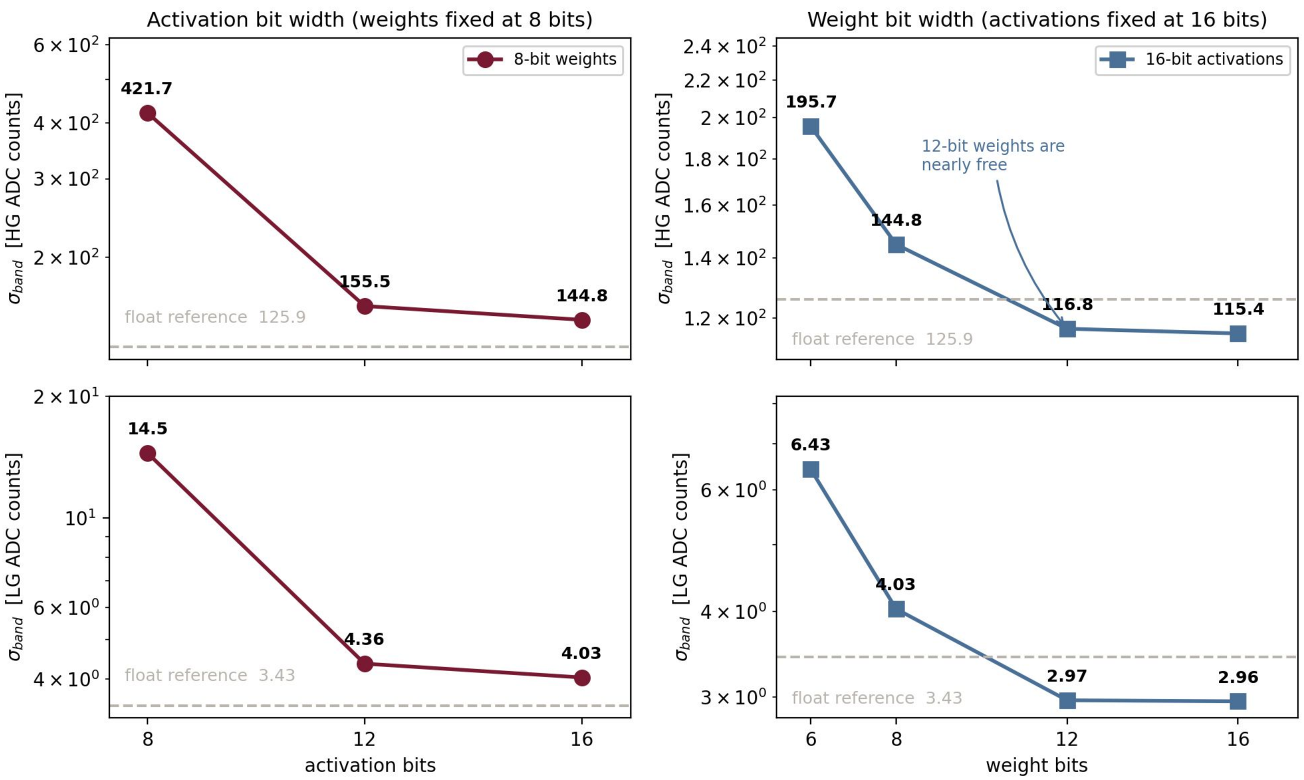
	a16	a12	a8
w16	115.4	128.5	429.4
w12	116.8	133.2	404.3
w8	144.8	155.5	421.7
w6	195.7	244.3	522.5

- Activations dominate. a8 is unusable at any weight width.
- Weights nearly free to 12 bits. w8 costs 29 HG ADC, w6 costs 80.

Why a8 fails. 256 output levels across 0 to 4095 LG ADC is a 16 LG ADC step, 640 HG ADC. That step is the low-energy banding.

Dead end. Four input transforms tried. None adds output levels and levels are what is short.

Resolution against quantization bit width, A1-A5, μ=200



From here on all sigma is 256-bin calibration, slide 5 is coarse. The band moves under 1 HG ADC between them, the global moves about 3.

Learnable bit-widths

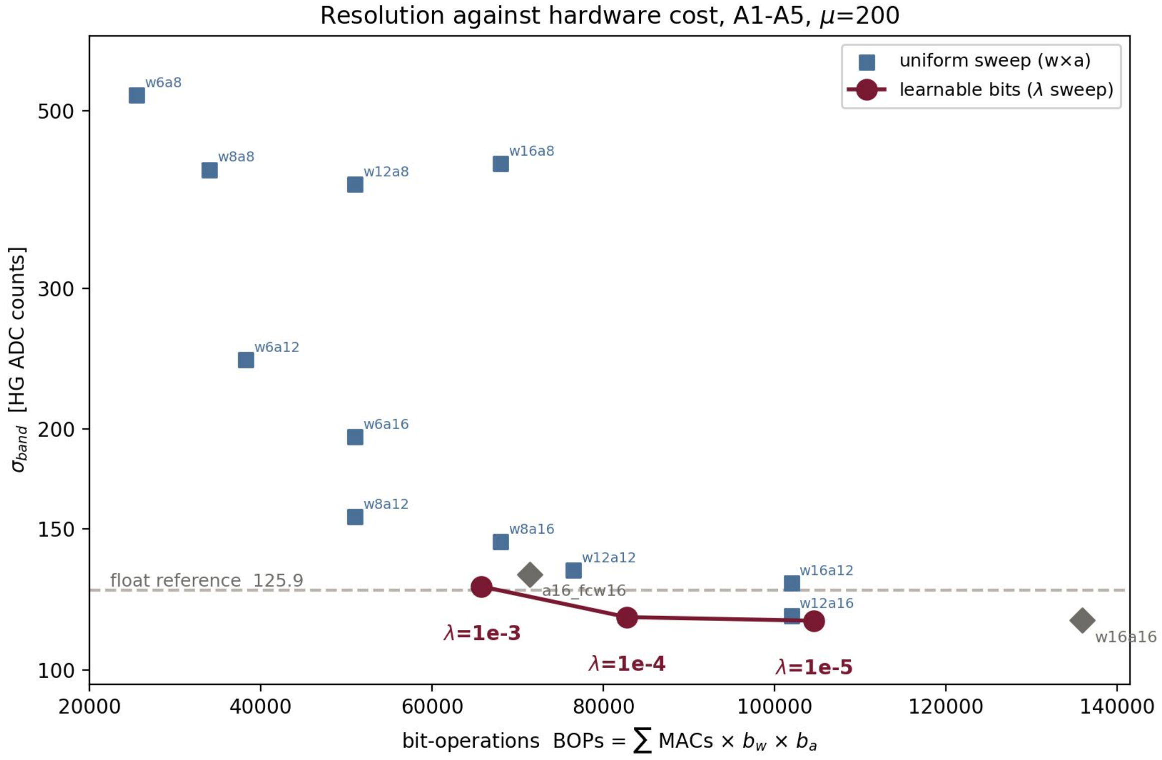
Bit width as a trained parameter, warm-started from w16/a16, penalty lambda trading bits against accuracy.

Learned at lambda = 1e-4, as weight bits / input-activation bits.

CONV1	CONV2	LINEAR
12/13	10/15	14/16

Model	bit ops	HG band (HG ADC)	LG band (LG ADC)
w16 / a16	135,936	115.4	2.96
w12 / a16, best uniform	101,952	116.8	2.97
learnable, lambda 1e-4	82,728	116.5	2.97

- 19% fewer bit operations than the best uniform point at the same band. 39% fewer than w16/a16 for 1% of band.
- The manual sweep only explores uniform pairs, so it cannot reach 12/13, 10/15, 14/16. The two methods are complementary.
- Pushed with a penalty above the task loss and a floor of 2 bits, it still stops at 9 to 13 bits and gives back the band rather than cross 8.



The quantized model beats the float model and why??

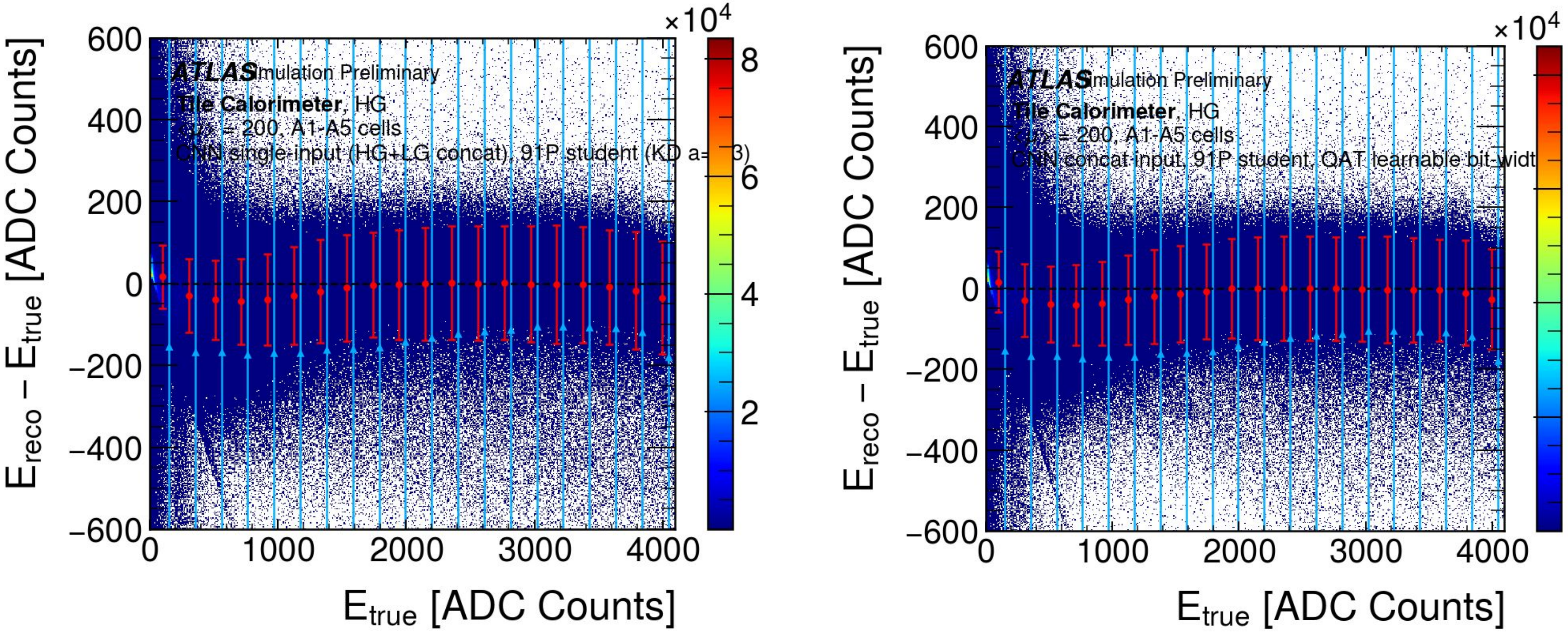
Model	HG global (HG ADC)	HG band (HG ADC)	LG band (LG ADC)
float student 91 P	91.1	125.9	3.43
QAT w16 / a16	87.7	115.4	2.96
QAT learnable bits	88.4	116.5	2.97

- Three candidates. The warm restart, quantization noise as a regulariser, or losing the distillation term since QAT trains on truth alone
- **Control run.** Continue the float model, same 100 epochs and learning rate, no quantization and no distillation.

It stayed at 127.3 HG ADC against the float student's 127.0. Restart and teacher removal are ruled out. The gain is in the quantization path.

Not pinned, which sub-cause inside that path. The claim stops with the control.

The quantized model beats the float model and why??



float against quantized HG band

Export to firmware and where the error came from

Brevitas → **QONNX** → **hls4ml** → **compile** → **predict**

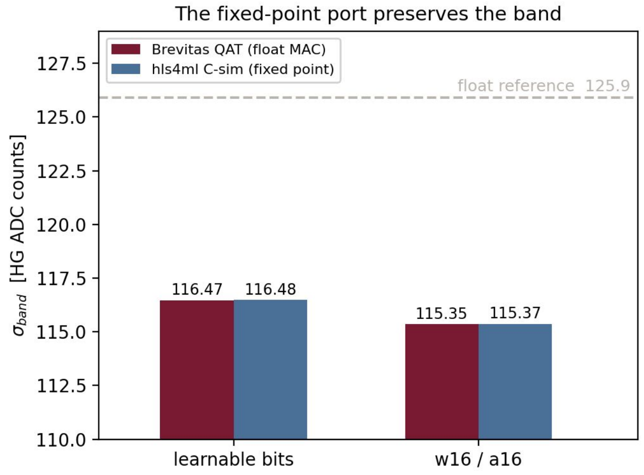
Predict runs the generated HLS C++ with the synthesised fixed-point types, so it is bit-accurate.

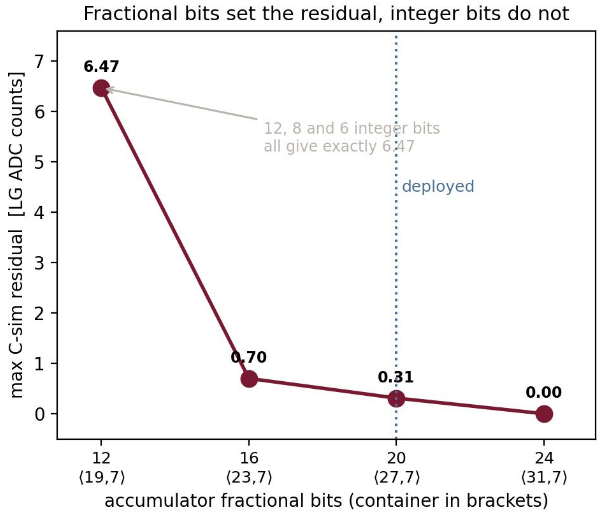
- LeakyReLU(0.25) survives as a 2^-2 literal and a ternary, not a lookup table. That was the one flagged export risk. Closed.
- The error was not the quantization. hls4ml sized the input word at 16 bits while our concatenated input needs 24, so samples were crushed before the input quantizer.
- Bisected each container with the other held wide, against a 48-bit reference. Integer bits nearly irrelevant, fractional bits and the accumulator are what matter.

Deployed. Input `fixed<20,2>`, accumulator `fixed<27,7>`, output `fixed<18,2>`, truncated after the compute so training is untouched.

Full test set, 10,659,229 events

Model	Brevitas HG band (HG ADC)	hls4ml C-sim HG band (HG ADC)	LG band (LG ADC)
learnable bits	116.47	116.48	2.97
w16 / a16	115.35	115.37	2.96





Brevitas and hls4ml bands overlaid

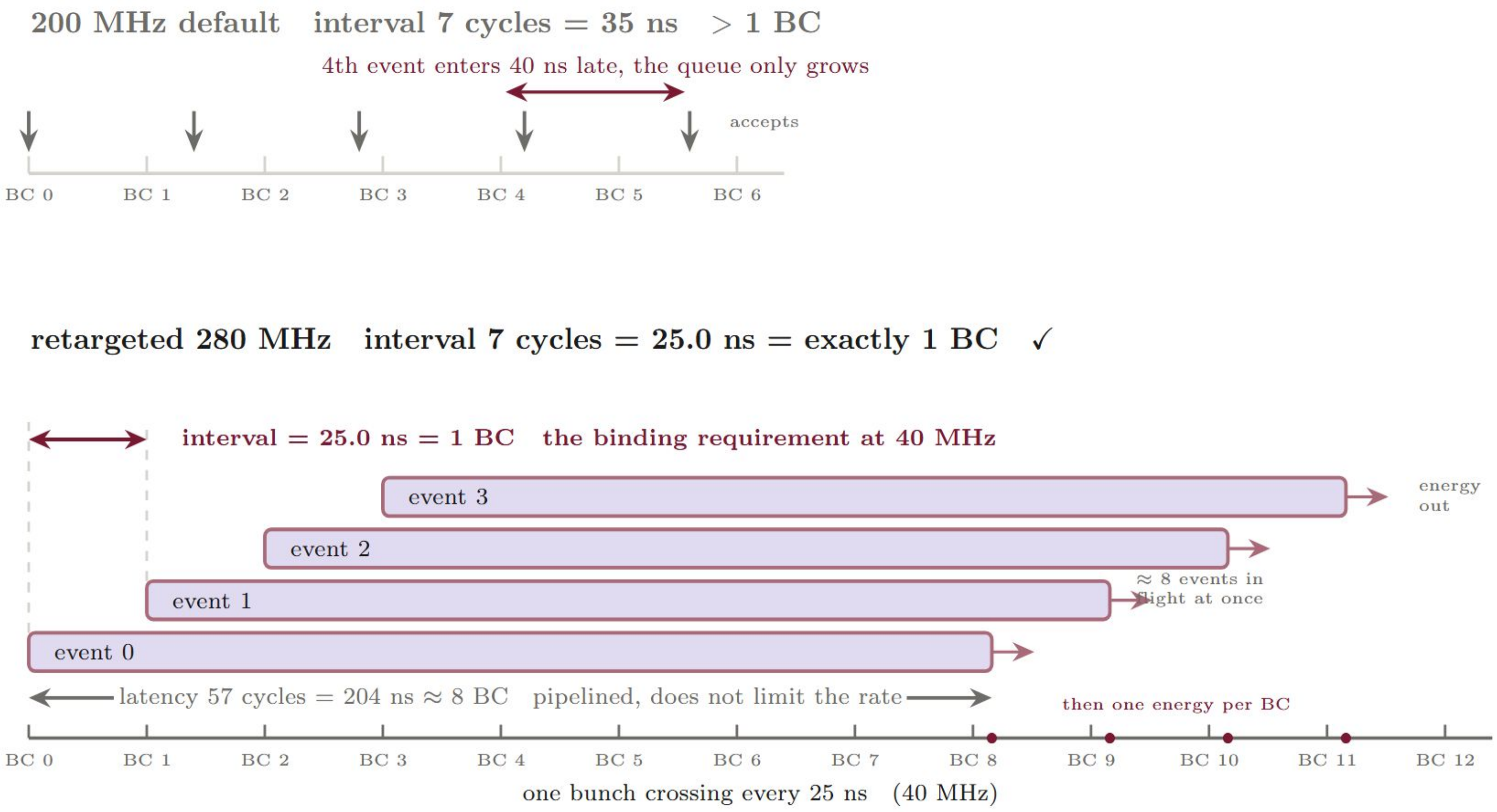
Meeting the bunch-crossing rate

Two numbers, used from here on. latency = input to result for one event. interval = how often a NEW event can be accepted. For 40 MHz the binding one is interval, at most 25 ns.

- The TilePPr block gets 225 ns, 9 bunch crossings and that is reconstruction PLUS sample delay. The 9-sample window takes 9 bunch crossings just to collect, so the network must pipeline inside it. It does not get 225 ns of compute.
- At 200 MHz the interval was 7 cycles = 35 ns, missing a bunch crossing. At 280 MHz, 7 cycles = 25.0 ns exactly.

Post-synthesis. 27,307 LUT (4.12%), 27,759 FF (2.09%), 80 DSP (1.45%), 0 BRAM. Latency 204 ns.

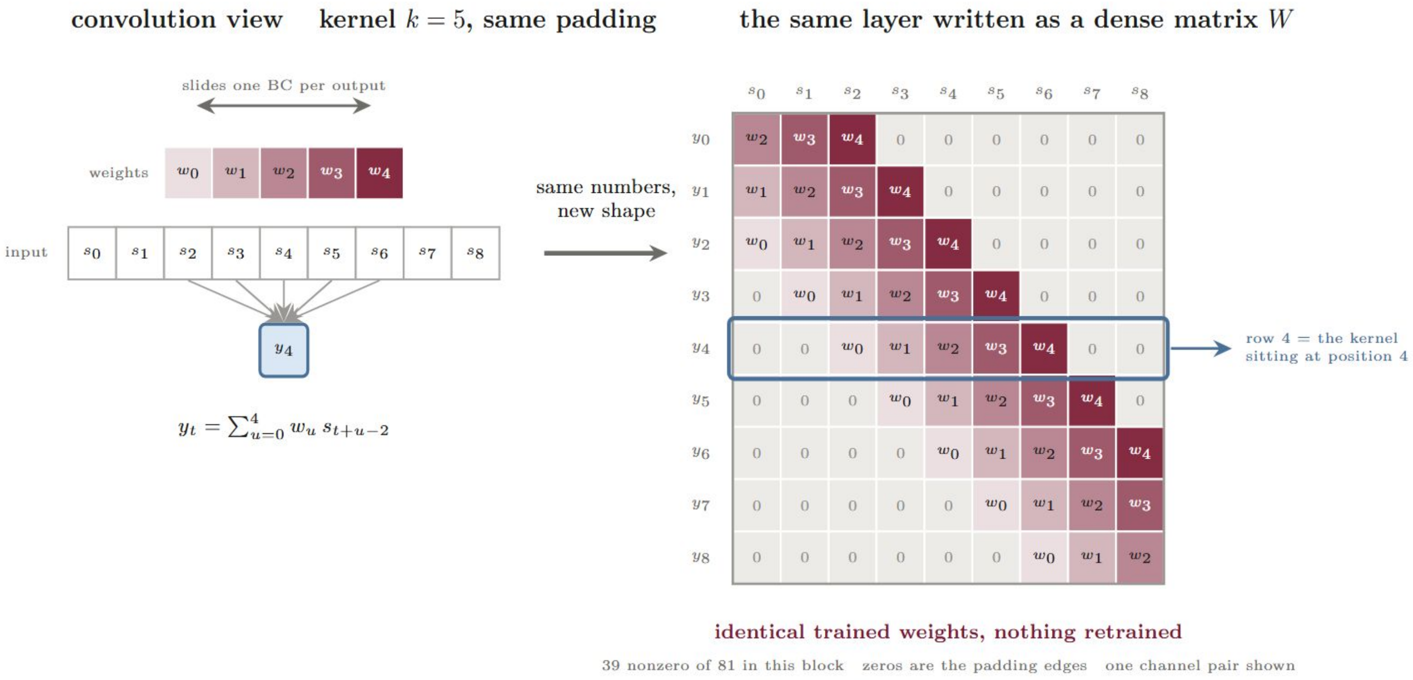
- Only the learnable-bit model reaches the rate. It is 7 cycles, w16/a16 needs 9 which means 360 MHz, not feasible here. The bit-operation saving cashes out as cycles, not LUTs.
- But one engine per channel means 77 engines, about 317% of device LUTs.



bunch-crossing clock against engine latency and interval

The dense rewrite to interval 1

- Everything except the two convolutions already ran at interval 1.
- A 9-sample same-padded convolution is a matrix multiply with fixed zeros. Both convolutions re-expressed as dense layers with block-Toeplitz weights. Same 91 parameters, no retraining, verified to 0.000000 LG ADC over 50,000 events before the converter will write.
- hls4ml's dense engine pipelines where its conv engine does not.



a 9-sample conv kernel redrawn as a block-Toeplitz matrix, same weights

Per engine at 280 MHz

	conv form	dense form
Interval	7 cycles = 25 ns	1 cycle = 3.571 ns
Channels served	1	7
Latency	204 ns	200 ns
LUT	27,307	21,502
FF	27,759	30,085
DSP	80	366

- Per channel it is cheaper both ways. LUT 27,307 to 3,100, DSP 80 to 52.
- DSP growth is throughput arithmetic. At interval 7 about seven multiplies shared one multiplier. At interval 1 they cannot. The 4.6x measures the sharing.

Does it actually route?

Synthesis does not check routing, so the design went through full out-of-context place-and-route. Out of context so I/O pad delays our block never sees are not charged to it.

DENSE FORM

WNS +0.073 ns

Single engine, dense, 280 MHz. All constraints met, WNS +0.073 ns, 0 failing of 69,914.

CONV FORM, SAME CLOCK

WNS -0.248 ns

The conv form at the same clock FAILS. WNS -0.248 ns, 610 failing of 61,027. The critical path is the time-sharing control machinery, so it is structural.

- The worst path is a wire, not logic. 96.5% routing on a fanout-217 net into the multiplier bank, zero logic levels. The 27% HLS clock uncertainty was absorbing exactly that.
- So the "no place-and-route yet" caveat was load-bearing and the dense rewrite is what makes 280 MHz closable at all, not just faster.

Does it actually route?

```
    WNS(ns)      TNS(ns)  TNS Failing Endpoints  TNS Total Endpoints      WHS(ns)      THS(ns)  THS Failing Endpoints
    -------      -------  ---------------------  --------------------     -------      -------  ---------------------
      0.073        0.000                      0                 69914       0.029        0.000                      0

    THS Failing Endpoints  THS Total Endpoints     WPWS(ns)     TPWS(ns)  TPWS Failing Endpoints  TPWS Total Endpoints
    ---------------------  -------------------     --------     --------  ----------------------  --------------------
                        0                69914        1.228        0.000                       0                 31740

All user specified timing constraints are met.
```

```
Slack (MET) :             0.073ns  (required time - arrival time)
  Source:                 layer54_out_V_4_reg_1107_reg[12]/C
                            (rising edge-triggered cell FDRE clocked by ap_clk  {rise@0.000ns fall@1.786ns period=3.571ns})
  Destination:            grp_dense_latency_ap_fixed_13_1_4_0_0_ap_fixed_27_7_4_0_0_config60_s_fu_63/mul_13s_8s_21_2_0_U122/dout_reg/DSP_A_B_DATA_INST/A[28]
                            (rising edge-triggered cell DSP_A_B_DATA clocked by ap_clk  {rise@0.000ns fall@1.786ns period=3.571ns})
  Path Group:             ap_clk
  Path Type:              Setup (Max at Slow Process Corner)
  Requirement:            3.571ns  (ap_clk rise@3.571ns - ap_clk rise@0.000ns)
  Data Path Delay:        3.195ns  (logic 0.113ns (3.537%)  route 3.082ns (96.463%))
  Logic Levels:           0
  Clock Uncertainty:      0.035ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 + PE
    Total System Jitter     (TSJ):    0.071ns
    Total Input Jitter      (TIJ):    0.000ns
    Discrete Jitter          (DJ):    0.000ns
    Phase Error              (PE):    0.000ns

    Location             Delay type                Incr(ns)  Path(ns)    Netlist Resource(s)
  -------------------------------------------------------------------    -------------------
                         (clock ap_clk rise edge)     0.000     0.000 r
                                                      0.000     0.000 r  ap_clk (IN)
                         net (fo=33242, unset)        0.000     0.000    ap_clk
    SLICE_X53Y379        FDRE                                         r  layer54_out_V_4_reg_1107_reg[12]/C
  -------------------------------------------------------------------    -------------------
    SLICE_X53Y379        FDRE (Prop_BFF_SLICEM_C_Q)
                                                      0.113     0.113 r  layer54_out_V_4_reg_1107_reg[12]/Q
                         net (fo=217, routed)         3.082     3.195    grp_dense_latency_ap_fixed_13_1_4_0_0_ap_fixed_27_7_4_0_0_config60_s_fu_63/mul_13s_8s_21_2_0_U122/dout_reg/A[28]
    DSP48E2_X5Y155       DSP_A_B_DATA                                 r  grp_dense_latency_ap_fixed_13_1_4_0_0_ap_fixed_27_7_4_0_0_config60_s_fu_63/mul_13s_8s_21_2_0_U122/dout_reg/DSP_A_B_DATA_INST/A[28]
```

worst-path timing report extract, routed dense engine

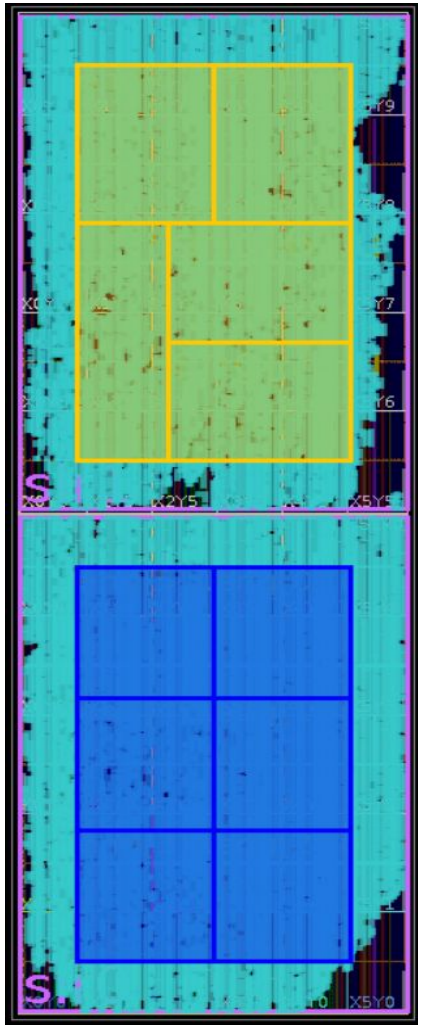
77 channels on one device

- 11 engines together, boundary-registered. 77 channels competing for one KU115.

Routed at 280 MHz

	value	of device
LUT	231,123	34.8%
FF	334,760	25.2%
DSP	4,026	72.9%
Worst slack	+0.012 ns	0 failing of 772,070

- DSP is exactly 11 x 366, no engine optimised away. Throughput fixes the multiplier count regardless of arrangement, 77 x 483 MACs x 40/280 = 5,313 against 5,520 on the device, so the 1,290 not spent as DSPs sit in LUT fabric, which 10 to 14 bit weights make cheap.
- The KU115 is two stacked dies joined by 17,280 wires. Unconstrained, 11 engines wanted 35,413 crossings and the placer refused. Pinned by hand, 6 and 5 per die, 2 pblocks in the P&R script, only the wrapper boundary crosses. 2 wires used.
- So quote per die, not per chip. Die 0 at 2,196 DSP (79.6%) and 81.1% of CLB sites, die 1 at 1,830 DSP (66.3%) and 69.5%. Site occupancy and LUT count are different measures and not directly comparable, an occupied CLB is not a full one, so the real headroom sits between the two.
- Left for the rest of TilePPr, about 19% of die 0, 30% of die 1, 27% of DSPs.



device floorplan, 11 engines pinned 6 and 5 across the two dies

Caveats. Margin is 12 picoseconds and structural, same path in two runs, so effort cannot widen it. One run landed at -0.014 ns after placement and physical optimization pulled it back, so that stays in the flow. Out-of-context idealises the clock network. This is our block alone, without the rest of the firmware.

Summary

- Knee near 1000 parameters. The 2005 P plateau is a capacity floor or a data ceiling, unresolved.
- Distillation at alpha = 0.3 buys 23% of band at 91 parameters, both gains, one output.
- Activations dominate quantization and 8 bits is not possible for this range. Learnable per-layer bits match the best uniform point at 19% fewer bit operations and are what makes the rate reachable.
- Firmware within 0.01 HG ADC of Brevitas on 10.66M events. Interval 1 at 280 MHz. 11 engines, 77 channels, placed and routed, all timing met.

Back to the firmware team, about 19% of die 0, 30% of die 1, 27% of DSPs.

NEXT

Pedestal correction, per channel in the orbit gap, then Athena

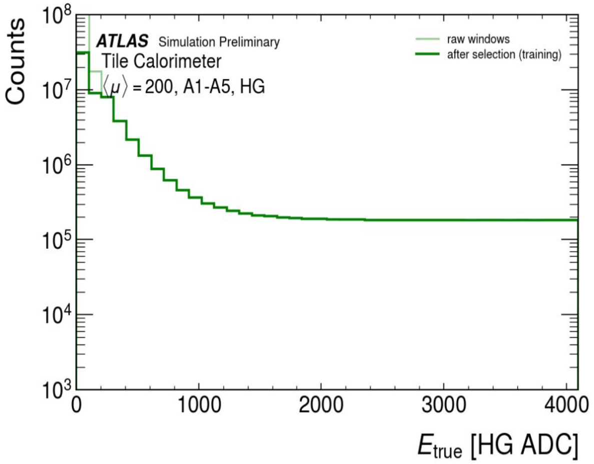
PARAMETERS	CHANNELS	CLOCK	INTERVAL
91	**77**	**280 MHz**	**1 cycle**

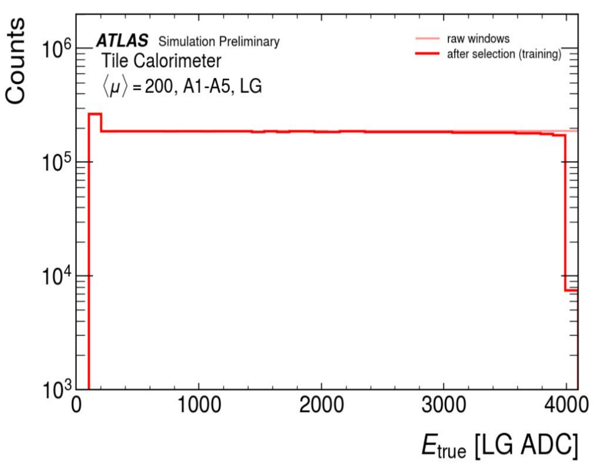
THANK YOU
QUESTIONS?

BACKUP

Backup, dataset verified values

quantity	value
bunch crossings	230,000
raw windows	294,164,480
kept	71,061,518 (24.16%)
split	70 / 15 / 15
train / val / test	49,743,062 / 10,659,227 / 10,659,229
test HG / LG	9,597,174 / 1,062,055
pedestal	exactly 100.0 counts, both gains
HG/LG ratio, ped-sub	40.26
median true energy	3.5 LG ADC
fraction in HG region	90.04%



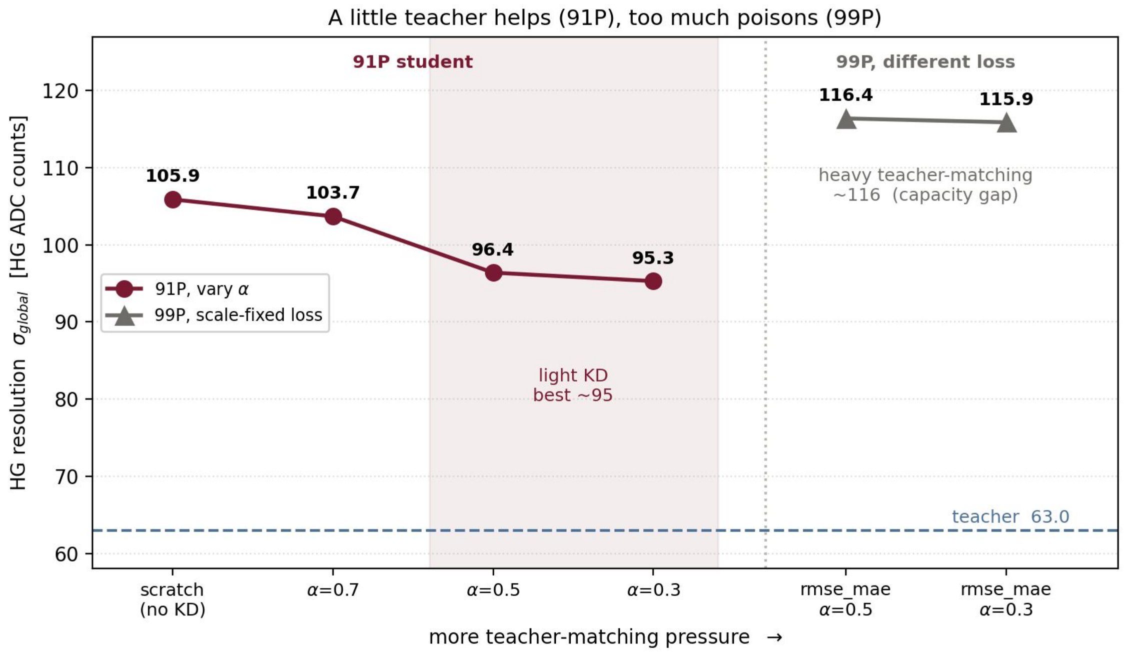


- The chain closes. 10,659,229 is exactly the event count in every sigma table in this deck.
- Earlier decks quoted 1 million bunch crossings, a 75 / 12.5 / 12.5 split and 95% in the lowest bin. Those came from a reference slide, not our files. No result changes.
- The directory holds 99 files and 1,390,000 bunch crossings. We use 16.5%.
- The sample is the 5%-signal high-gain-biased production, chosen deliberately. More energetic signals than minimum bias, tilted to the harder high-gain region.

Backup, full sub-100-parameter student matrix

Coarse HG binning

Run	params	HG global (HG ADC)	HG band (HG ADC)	LG band (LG ADC)
from scratch	91	105.9	164.8	4.72
alpha 0.7	91	103.7	148.3	4.14
alpha 0.5	91	96.4	128.5	3.89
alpha 0.3, best	**91**	**95.3**	**127.0**	**3.43**
alpha 0.3	99	106.0	149.6	4.84
alpha 0.3	106	98.1	125.7	3.76
alpha 0.3, seed 2	106	97.7	139.6	3.77
alpha 0.1, teacher 533 P	99	104.8	138.4	4.17
alpha 0.1, teacher 1005 P	99	96.6	148.2	4.50
scale-fixed term, alpha 0.3	99	115.9	194.4	5.90
scale-fixed term, alpha 0.5	99	116.4	184.4	5.73
k-teacher, 5 teachers	91	102.3	145.2	3.97
intermediate-layer feature	91	105.8	152.6	4.71
teacher-assistant 441 to 91	91	101.3	134.1	4.54
relative-resolution loss	91	109.8	212.6	7.50
teacher, anchor	1005	63.0	85.8	2.70



- Loss is L = alpha * L_true + (1 - alpha) * L_KD. No temperature, that is a classification construct and this is regression.
- The band is seed-noisy here. Two 106 P seeds give 125.7 and 139.6, a 14 HG ADC swing, so 106 P's apparent edge sits inside noise. 91 P is chosen on the firmware budget.

Backup, learnable bit-width penalty scan

lambda	learned bits, w/a per layer	bit ops	HG band (HG ADC)	LG band (LG ADC)
1e-3	10/13, 9/13, 11/15	65,763	127.2	3.37
1e-4	12/13, 10/15, 14/16	82,728	116.5	2.97
1e-5	13/15, 12/16, 15/18	104,598	115.3	2.94

- 1e-4 is the deliverable. 1e-5 is marginally better but not deployable, one activation drifts up to 18 bits past our 16-bit ceiling. The method has no maximum-width constraint.
- 1e-3 lands at exactly the float band, 127.2 against 127.0. Squeezed hard the network gives back precisely the distillation gain and no more.
- Nothing goes below 9 bits anywhere, even at 1e-3 where the penalty starts at twice the task loss and the available floor is 2 bits.

Backup, container bisection

2,000-event residual in LG ADC

input word	accumulator	max	mean
fixed<48,16>	fixed<48,16>	0.00	0.00
fixed<16,8>	fixed<48,16>	16.94	4.62
fixed<48,16>	fixed<16,8>	91.40	31.18
fixed<48,16>	fixed<24,12>	6.47	1.37
fixed<48,16>	fixed<23,7>	0.70	0.14
fixed<48,16>	fixed<27,7>	0.31	0.02

- Integer bits are irrelevant. 12, 8 and 6 integer bits all give exactly 6.47. Fractional bits set the residual and the accumulator dominates.
- Each container was bisected with the other held wide, then confirmed with a full-band run.
- Everything runs on the normalised target, so 7 signed integer bits give 64x headroom on a quantity never exceeding 1. The x4095 rescale downstream gives a 16-bit word at 1/16 LG ADC.

Backup, the two-die placement problem

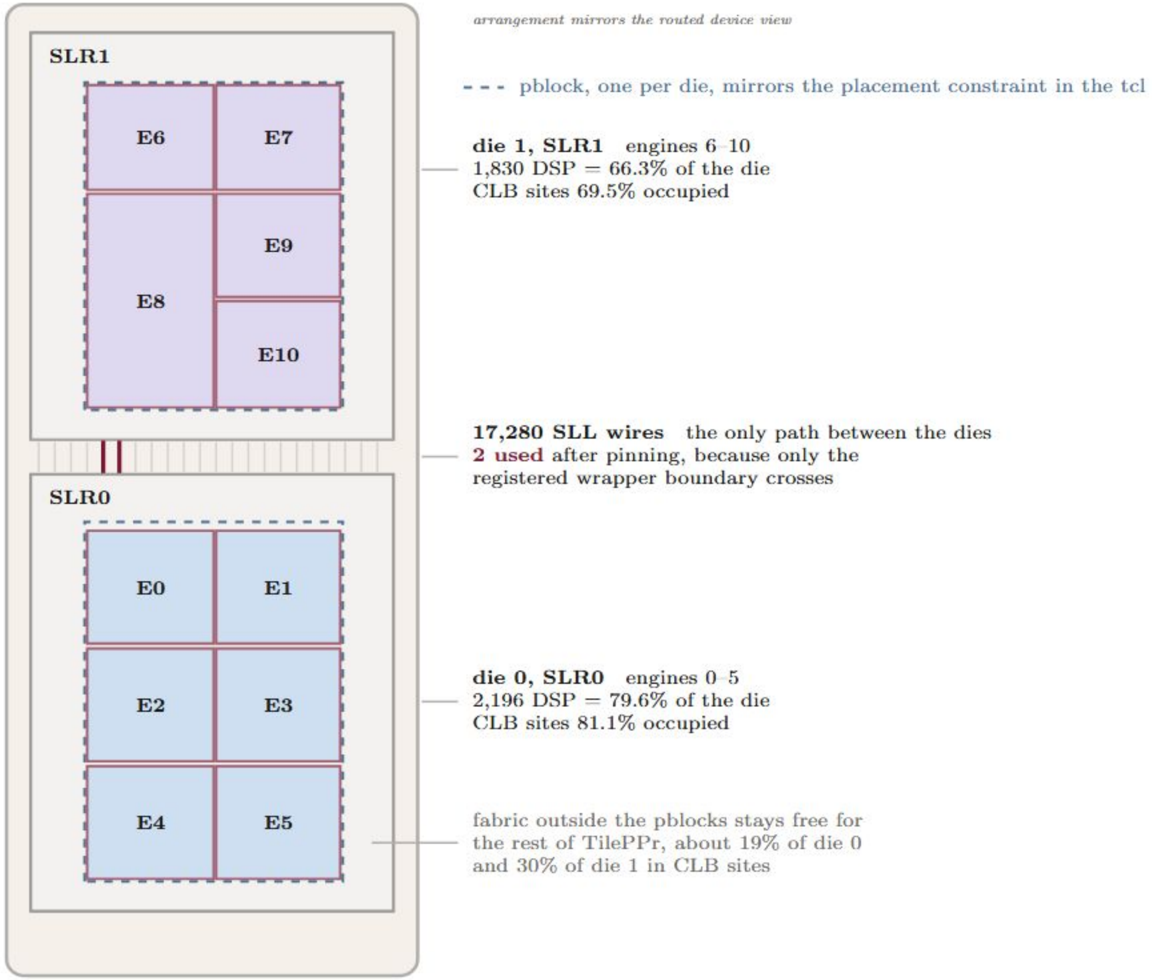
- The KU115 is a stacked-silicon part. Two dies on an interposer, joined by exactly 17,280 SLL wires, which are the only path between them.

- Why unconstrained placement was so expensive??? An engine placed across the boundary sends its own internal nets over the join and engine internals are densely wired, one net alone has a fanout of 217 into the multiplier bank. Eleven engines straddling the join asked for 35,413 crossings against 17,280 available and the placer stopped with an error.

- Per-die DSP capacity is about 2,760, so at 366 DSP per engine no die holds more than seven. Eleven engines must split and 6 + 5 is the balanced choice.

- After pinning, only the registered wrapper boundary crosses. 2 SLL wires used, down from 35,413 demanded, four orders of magnitude from one floorplan constraint.

XCKU115, one FPGA, two stacked dies



× unconstrained placement demanded 35,413 crossings, more than the 17,280 available, the placer refused

✓ engines pinned 6 + 5 per die, 2 SLLs used, all timing met at 280 MHz

each engine serves 7 channels • 27% of the chip's DSPs remain for the rest of the firmware

Backup, the reference models and why the comparison is not like-for-like

Model	params	HG sigma_avg_err	LG sigma_avg_err
reference CNN	147	72.12	8.36
reference MLP	148	99.76	10.75

- Three reasons no comparison is claimed here.
  1. Different metric. sigma_avg_err is a global std, ours is the band. Not comparable.
  2. Different data and pipeline and a global sigma depends on the injection distribution.
  3. It is not two per-gain networks. It is one model on a gain-mixed input, scored on the high-gain and LG-saturated subsets separately, so it needs the same instance count we do.
- The real difference is architectural, not a score. Our concatenated input keeps both gains losslessly, where a gain-mixing rule keeps one and discards the other.
- Doing it properly means re-running both on a common sample with a common metric.

Backup, architecture, filters and methodology

Teacher, Conv1d(1->18,k5) -> Conv1d(18->14,k3) -> Linear(126->1), 1005 parameters.

Student, Conv1d(1->4,k5) -> Conv1d(4->3,k3) -> Linear(27->1), 91 parameters.

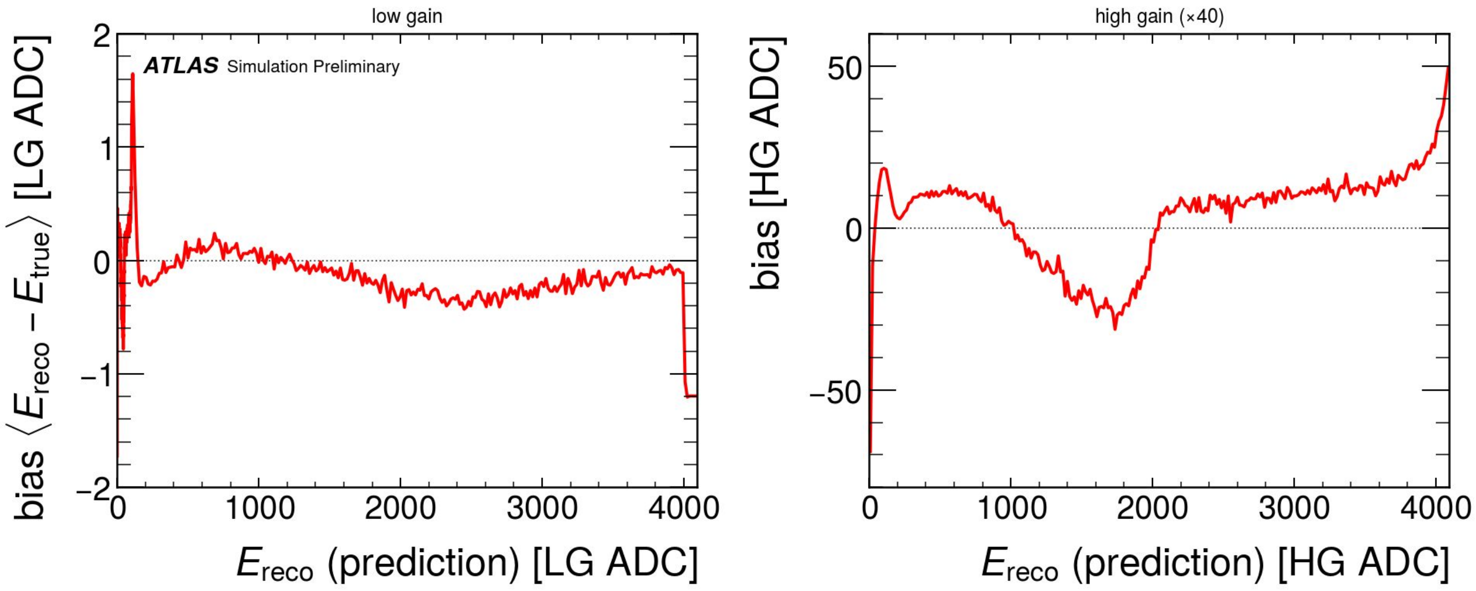
- Kernels are physics-driven. k5 then k3 gives a 7-bunch-crossing receptive field, enough to see out-of-time pile-up at the window edges. Two k3 layers reach only 5.

Filters. Drop the window if any sample saturates low gain at 4095 counts, if the central target is out of range, or randomly half the windows below 5.0 LG ADC as class balancing.

Normalisation. Input divided by 2^24 - 1, target by 4095. Different scales, both stored.

- Calibration is a function of the PREDICTION, not true energy, which is unavailable at reconstruction time. One curve fixes both gains, high gain binned at 256 points.

- A low-energy HG dip survives calibration and does not move as the binning goes 6 to 40 to 256, so it is not coarse-bin under-correction. Hypothesis, regression-to-the-mean, the bias lives in true-energy space. A hypothesis, not a result.



the bias curve we subtract, low gain and the x40 high-gain view

Backup, what is not tested

- The 11-engine result is our block alone on an empty device. Congestion against the rest of the TilePPr firmware is untested and cannot be tested from here.
- Out-of-context idealises the clock network, so the integrated build adds real skew. Final signoff belongs to the firmware build.
- Vitis HLS 2022.1, one release below hls4ml's floor of 2022.2. The only error is an unsupported array-partition command on 36 and 27 element arrays and co-simulation passes. Re-check on 2022.2 before quoting formally.
- Co-simulation verified function, RTL against the C model on 100 real events. It did not measure the interval, which comes from the synthesis schedule.
- The 280 MHz margin is 12 picoseconds and structural. Met in every run, not comfortable.